SI 507 Final Project Overview & Proposal Guidelines

SI 507 Teaching Team

1 Introduction

SI 507 has equipped you with the fundamental primitives of modern software: object-oriented design, graph and tree structures, algorithms, and APIs. These are the exact building blocks of the agentic systems reshaping how software gets built and used. Graphs model information flow. Trees structure decision-making. APIs connect systems to the world. Object-oriented design gives complex programs the architecture to do something meaningful.

Your final project is an opportunity to assemble these tools into something ambitious and original. AI tools are available to you as collaborators in the building process — using them reflects the current reality of software development. You may also build a project where an LLM is part of the product itself, powering a feature or interaction that would be impossible otherwise. The standard we hold you to is this: you understand every part of what you built, and your tests demonstrate that understanding.

2 Core Requirements

2.1 Graph or Tree Structure (Required)

Your project must use a graph or tree as its primary organizational framework. The world is full of networks waiting to be found: social connections, geographic relationships, recommendation chains, decision hierarchies, dependency structures. Your job is to find the network hiding in your domain and build around it.

Clearly define what your **nodes** represent and what your **edges** represent. The most interesting projects are those where the structure itself is the insight — where traversing the graph reveals something you could not see by looking at a table of data.

2.2 Object-Oriented Design (Required)

Your project must demonstrate thoughtful object-oriented programming. At minimum:

- Define at least one meaningful class that represents a concept in your domain
- Use class attributes and methods purposefully
- Show relationships between objects where appropriate
- Write modular, readable, well-documented code

2.3 Real Data (Required)

Work with real-world data from genuine sources: APIs, web scraping, CSV files, or databases. Synthetic data is permitted only with explicit justification in your proposal. Demonstrate appropriate complexity through either the volume of data processed or the difficulty of data access.

2.4 Four Modes of Interaction (Required)

Your program must provide at least four distinct ways for users to explore the data. Examples include search and query, detailed node views, rankings and comparisons, relationship or path finding, filtering, and external links.

2.5 Interface

A text-based command-line interface is acceptable and will receive full credit for the interface component. Building a **Streamlit app** is strongly encouraged. Streamlit is the kind of tool you will encounter in data science and industry settings, and a well-built Streamlit interface will make your project genuinely portfolio-worthy.

2.6 Testing (Required)

Your test suite is the proof that you understand what you built. Tests must go beyond confirming that code runs — they should verify meaningful behaviors, handle edge cases, and collectively paint a picture of what your program does. Reading your tests should tell someone unfamiliar with your project what it is supposed to do.

3 Example Projects

The following examples are meant to illustrate not just *what* to build, but *how to think* about going beyond a basic implementation. The question to keep asking is: what does the graph reveal that a simple list or table cannot?

3.1 Example 1: NBA Teammate Network

The Kevin Bacon number problem from class translates naturally to NBA players: build a graph where players are nodes and edges connect players who were teammates. But the interesting questions emerge from what you layer on top of that structure.

- Which players are most connected *across eras* — the Kevin Bacons of the NBA? Does centrality in the teammate graph correlate with All-Star selections, or does it just reflect longevity?
- Weight edges by the number of seasons two players shared. Does playing together longer create tighter clusters?
- Incorporate hometown data. Do players from the same city or region end up as teammates more often than chance would predict?
- Trace how a single superstar trade restructures the entire network. What does the graph look like before and after LeBron James moves teams?
- Use an LLM to generate a natural-language narrative of the shortest path between two players, explaining each connection in context.

Data sources to explore: basketball-reference.com CSV exports, the NBA API, Wikipedia infoboxes via scraping.

3.2 Example 2: Restaurant Recommender

A restaurant recommendation system is a classic project, but replicating a subset of Yelp is not the goal. The goal is to find the insight that Yelp does not offer.

- Build a neighborhood graph where restaurants are nodes and edges connect places that share a meaningful relationship: similar cuisine, overlapping reviewer populations, or geographic proximity. What does your city's food scene look like as a network?
- Integrate a second dataset — weather data, neighborhood demographic data, or event listings. What should you eat near a concert venue on a rainy Friday?
- Use Spotify or Last.fm data to connect restaurant vibes to music moods. Recommend a dinner spot that matches a playlist.
- Use an LLM to generate a "dining narrative" from a path through the graph: a story of an evening that moves from cocktails to dinner to dessert, each stop connected to the last.

Data sources to explore: Yelp Fusion API, Google Places API, OpenWeatherMap, Spotify API.

4 Project Milestones

Total Project Points: 200

- Proposal: 40 points (completion grade, banked before final submission)
- Checkpoint: 20 points (completion grade, banked before final submission)
- Final Submission: 140 points (graded on rubric below)

4.1 Milestone 1: Project Proposal — 40 points

Due: March 20

Submit a proposal of 0.5–1 page describing one project idea. You may include additional ideas if you want feedback on alternatives, but one well-developed idea is sufficient. Your proposal will receive iterative feedback and must be approved before you proceed. You may resubmit as many times as needed until it is approved.

Your proposal should address the following:

- **Target Grade:** What grade are you aiming for on the final project? Refer to Section **??** and state which tier you are targeting. This is a commitment, not a ceiling — it helps us give you the right feedback and helps you plan your scope accordingly. If your circumstances change, reach out to the teaching team.
- **Domain and Dataset:** What is your topic? What data sources do you plan to use, and do you have reason to believe they are accessible in the format you expect?
- **Network Structure:** What will your nodes represent? What will your edges represent? Why is a graph or tree the right structure for this domain?
- **Dream Scope:** If time and resources were unlimited, what would you build?
- **Realistic Scope:** What can you realistically deliver by April 24, given the grade you are targeting?

Grading: Completion grade. Full points awarded for a genuine, thoughtful proposal. Resubmit based on feedback until approved.

4.2 Milestone 2: Project Checkpoint — 20 points

Due: April 3

Submit one of the following two statements:

- **Statement A:** "I have verified that the data I plan to use is accessible in the format I described in my proposal, and I am proceeding as planned."
- **Statement B:** "I have encountered an issue with my data or project direction and will follow up with the teaching team via email or Piazza."

Grading: Completion grade. Full points for submitting either a confirmation or a genuine question.

4.3 Milestone 3: Final Submission — 140 points

Due: April 24

See the grading rubric below for full details.

5 Grading Rubric

5.1 Category Tiers

Each of the five categories below is assessed at one of three levels: **Minimum**, **Strong**, or **Exceptional**.

1. Data & Architecture

- *Minimum:* One real dataset; graph or tree structure is present and used for at least one interaction
- *Strong:* Multiple datasets integrated meaningfully, or data access involves genuine complexity (authenticated APIs, scraping, significant cleaning)
- *Exceptional:* The network structure reveals something about the domain that would be invisible in a flat dataset — the graph is the insight

2. Object-Oriented Design

- *Minimum:* At least one meaningful class with attributes and methods; basic docstrings present
- *Strong:* Multiple classes with thoughtful relationships; clean separation of concerns; consistent style throughout
- *Exceptional:* Code is readable enough that someone unfamiliar with the project could extend it; design choices are clearly intentional

3. Interactivity & Interface

- *Minimum:* Four distinct interaction modes; command-line interface acceptable
- *Strong:* Streamlit app with clean layout and intuitive navigation
- *Exceptional:* Interface feels like a product — someone outside this class would find it useful or compelling

4. Testing

- *Minimum:* A test suite exists and passes; tests cover core functionality
- *Strong:* Tests cover meaningful behaviors beyond happy paths; edge cases are addressed

- *Exceptional:* Reading the test suite tells you what the program does — tests serve as living documentation of the system's behavior

5. Ambition & Insight

- *Minimum:* Project delivers on its proposal; scope is appropriate for the timeline
- *Strong:* Project integrates an LLM as a development collaborator or as a product feature; or scope meaningfully exceeds expectations
- *Exceptional:* Something genuinely surprising — a dataset, use case, or capability that makes the grader stop and think

5.2 Grade Mapping

You choose your grade. Declare your target in your proposal, and build to that standard. The teaching team will calibrate feedback to match your stated ambition.

Target Grade	Requirement
A	Minimum in all 5 categories, and Strong or Exceptional in at least 4. A project that is off-the-charts Exceptional in a single category may compensate for being at Minimum in one other.
A-	Minimum in all 5 categories, and Strong or Exceptional in at least 2.
B	Minimum in all 5 categories.

Targeting below a B? If your circumstances make a B feel out of reach, reach out to the teaching team in your proposal. We will work with you to figure out what makes sense.

6 Getting Help

- Start early and come to office hours
- Post questions on Piazza — especially if your data source turns out to be inaccessible
- Iterate on your proposal based on feedback; all proposals must be approved before you proceed
- Remember: this is portfolio-worthy work. Treat it like something you would be proud to show a future employer or research supervisor.